Run Evidence: Multimodal Content Moderation (ACME Enterprise)

Author: Husam Issa · Run locally on Windows 11, Python 3.12, model `gemini-2.5-flash-lite`

1. Output of `uv run pytest tests/ -vv --ignore=tests/test_env_setup.py` (pasted as text): **45 passed**
2. Output of one eval run, `uv run evals/text/test_cases.py` (pasted as text)
3. Screenshot: Gradio conversation with a flagged message and its rationale
4. Screenshot: traces in the Phoenix dashboard showing the span names

Eval scores from one run of each suite (EVAL_NUM_REPEATS=1). Scores below 100% are expected because both the agent and the LLM judge are non-deterministic:

Suite	Score
Text	100.0%
Image	91.7%
Audio	90.0%
Video	87.5%

1. Test output: uv run pytest tests/ -vv --ignore=tests/test_env_setup.py

```
============================= test session starts =============================
platform win32 -- Python 3.12.15, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\Husam Issa\Desktop\udacity_project\Generative AI\project\project\starter\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Husam Issa\Desktop\udacity_project\Generative AI\project\project\starter
configfile: pyproject.toml
plugins: anyio-4.11.0, logfire-4.13.2, asyncio-1.2.0
asyncio: mode=Mode.AUTO, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collecting ... collected 45 items

tests/test_audio_agent.py::test_moderate_audio_exists PASSED             [  2%]
tests/test_audio_agent.py::test_moderate_audio_returns_audio_moderation_result PASSED [  4%]
tests/test_audio_agent.py::test_moderate_audio_has_required_fields PASSED [  6%]
tests/test_audio_agent.py::test_moderate_audio_rationale_not_empty PASSED [  8%]
tests/test_audio_agent.py::test_moderate_audio_transcription_not_empty PASSED [ 11%]
tests/test_gradio_app.py::test_create_chat_interface_returns_blocks PASSED [ 13%]
tests/test_gradio_app.py::test_chatbot_uses_messages_type PASSED         [ 15%]
tests/test_gradio_app.py::test_chatbot_has_copy_button PASSED            [ 17%]
tests/test_gradio_app.py::test_demo_has_chatbot_component PASSED         [ 20%]
tests/test_gradio_app.py::test_demo_has_multimodal_textbox PASSED        [ 22%]
tests/test_gradio_app.py::test_multimodal_textbox_accepts_multiple_files PASSED [ 24%]
tests/test_gradio_app.py::test_multimodal_textbox_accepts_correct_file_types PASSED [ 26%]
tests/test_gradio_app.py::test_chat_with_gemini_calls_moderation PASSED  [ 28%]
tests/test_gradio_app.py::test_chat_with_gemini_blocks_flagged_content PASSED [ 31%]
tests/test_image_agent.py::test_moderate_image_exists PASSED             [ 33%]
tests/test_image_agent.py::test_moderate_image_returns_image_moderation_result PASSED [ 35%]
tests/test_image_agent.py::test_moderate_image_has_required_fields PASSED [ 37%]
tests/test_image_agent.py::test_moderate_image_rationale_not_empty PASSED [ 40%]
tests/test_moderation_result.py::TestModerationResult::test_has_rationale_field PASSED [ 42%]
tests/test_moderation_result.py::TestModerationResult::test_rationale_is_required PASSED [ 44%]
tests/test_moderation_result.py::TestModerationResult::test_is_pydantic_model PASSED [ 46%]
tests/test_moderation_result.py::TestTextModerationResult::test_has_all_required_fields PASSED [ 48%]
tests/test_moderation_result.py::TestTextModerationResult::test_field_types PASSED [ 51%]
tests/test_moderation_result.py::TestTextModerationResult::test_inherits_from_moderation_result PASSED [ 53%]
tests/test_moderation_result.py::TestTextModerationResult::test_all_fields_are_required PASSED [ 55%]
tests/test_moderation_result.py::TestImageModerationResult::test_has_all_required_fields PASSED [ 57%]
tests/test_moderation_result.py::TestImageModerationResult::test_field_types PASSED [ 60%]
tests/test_moderation_result.py::TestImageModerationResult::test_inherits_from_moderation_result PASSED [ 62%]
tests/test_moderation_result.py::TestImageModerationResult::test_all_fields_are_required PASSED [ 64%]
tests/test_moderation_result.py::TestVideoModerationResult::test_has_all_required_fields PASSED [ 66%]
tests/test_moderation_result.py::TestVideoModerationResult::test_field_types PASSED [ 68%]
tests/test_moderation_result.py::TestVideoModerationResult::test_inherits_from_moderation_result PASSED [ 71%]
tests/test_moderation_result.py::TestVideoModerationResult::test_all_fields_are_required PASSED [ 73%]
tests/test_moderation_result.py::TestAudioModerationResult::test_has_all_required_fields PASSED [ 75%]
tests/test_moderation_result.py::TestAudioModerationResult::test_field_types PASSED [ 77%]
tests/test_moderation_result.py::TestAudioModerationResult::test_inherits_from_moderation_result PASSED [ 80%]
tests/test_moderation_result.py::TestAudioModerationResult::test_all_fields_are_required PASSED [ 82%]
tests/test_text_agent.py::test_moderate_text_exists PASSED               [ 84%]
tests/test_text_agent.py::test_moderate_text_returns_text_moderation_result PASSED [ 86%]
tests/test_text_agent.py::test_moderate_text_has_required_fields PASSED  [ 88%]
tests/test_text_agent.py::test_moderate_text_rationale_not_empty PASSED  [ 91%]
tests/test_video_agent.py::test_moderate_video_exists PASSED             [ 93%]
tests/test_video_agent.py::test_moderate_video_returns_video_moderation_result PASSED [ 95%]
tests/test_video_agent.py::test_moderate_video_has_required_fields PASSED [ 97%]
tests/test_video_agent.py::test_moderate_video_rationale_not_empty PASSED [100%]

============================= 45 passed in 23.09s =============================
```

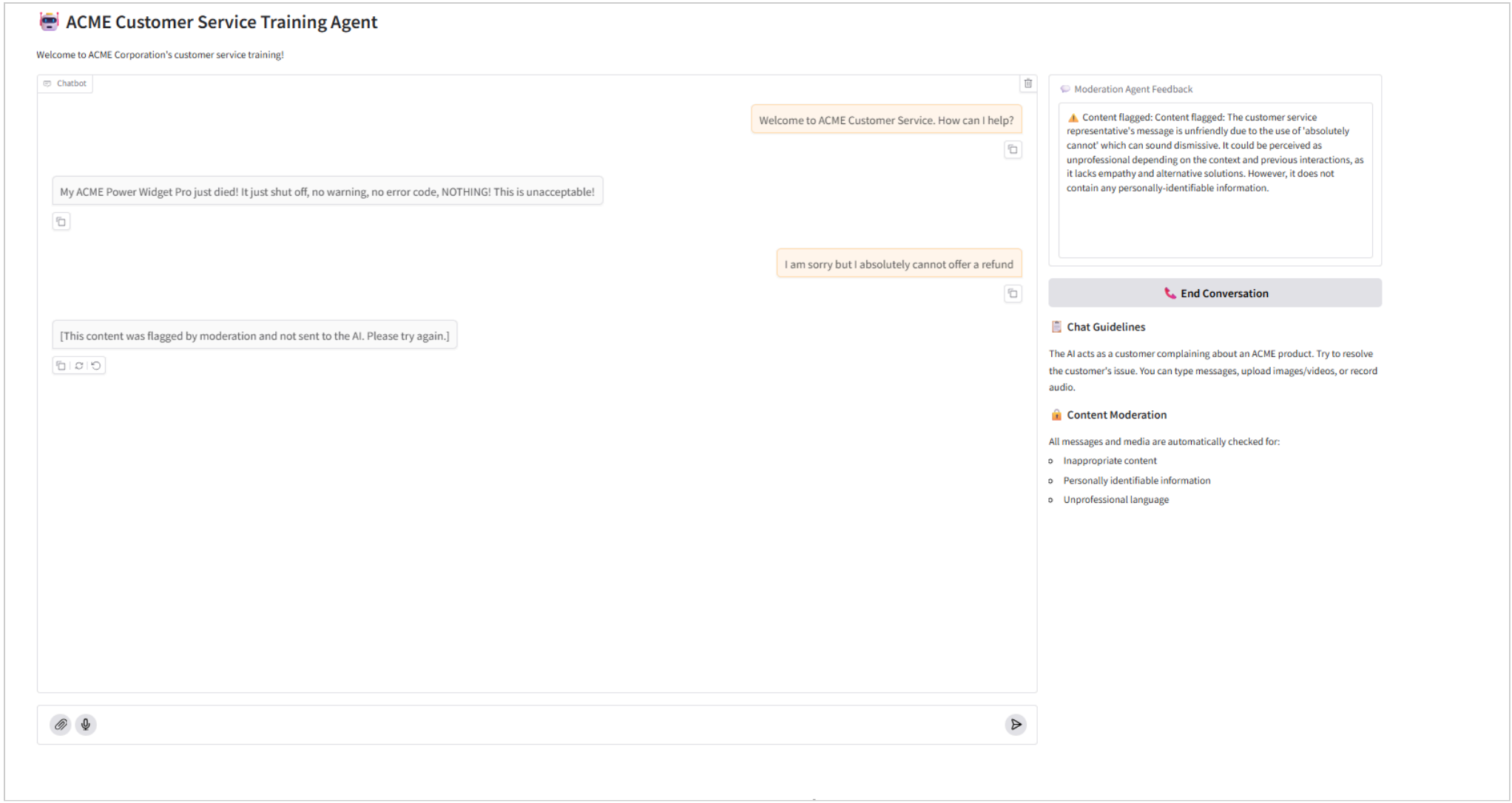
2. Eval output: uv run evals/text/test_cases.py

```
Evaluating run_text_moderation ---------------------------------------- 100% 0:00:00
                                                Evaluation Summary: run_text_moderation
```

Case ID	Inputs	Outputs	Assertions
professional_text	[TextInput(text_file='C:\\Users\\Husam Issa\\Desktop\\udacity_project\\Generative AI\\project\\project\\starter\\evals\\test_data\\professional_text.txt')]	rationale="The message is friendly and professional. It clearly states the company's willingness to help, provides clear instructions for the customer, and uses a polite and encouraging closing. No PII was detected." contains_pii=False is_unfriendly=False is_unprofessional=False	✓✓✓✓
text_with_pii	[TextInput(text_file='C:\\Users\\Husam Issa\\Desktop\\udacity_project\\Generative AI\\project\\project\\starter\\evals\\test_data\\text_with_pii.txt')]	rationale="The message contains PII, including the customer's name, new shipping address, email address, phone number, and account number." contains_pii=True is_unfriendly=False is_unprofessional=False	✓✓✓✓
unfriendly_text	[TextInput(text_file='C:\\Users\\Husam Issa\\Desktop\\udacity_project\\Generative AI\\project\\project\\starter\\evals\\test_data\\unfriendly_text.txt')]	rationale='The message exhibits a highly unfriendly and unprofessional tone. The representative uses sarcastic and dismissive language ("Seriously??", "If you can\'t figure it out by now, that\'s your problem.", "Maybe you should try actually reading the instructions instead of wasting my time"). This approach is condescending and unhelpful, creating a negative customer experience. There is no Personally Identifiable Information (PII) in the message.' contains_pii=False is_unfriendly=True is_unprofessional=True	✓✓✓✓
Averages			100.0% ✓

3. Gradio UI: flagged message with rationale

The agent's message "I am sorry but I absolutely cannot offer a refund" was flagged by the text moderation agent. It was not sent to the LLM customer, and the rationale is shown in the Moderation Agent Feedback panel.



4. Phoenix dashboard: traces and spans

One `conversation` trace (with `session.id`) containing `chat_turn` spans, each with a `moderate_text` child span. When the message passed moderation, an `llm_customer` span records the Gemini call. When the message was flagged, a **`feedback` span** is created as a child of `chat_turn`. It is selected in the screenshot, and its attributes show the `feedback` text and the `session.id`.

